

Semi-Diophantine Equation Solutions

2021

Abstract

For my teacher: a semi-Diophantine of the form $x^n + y^n = z$ has a solution

1 General Form

1.1 Natural Numbers

Consider the following equation,

$$x^n + y^n = z$$

and $Q = [x, y]$ where $x, y, z, n \in \mathbb{N}$ and we have to find the unknown n . Without loss of generality let's consider $x > y$, then $n \approx \log_x(z)$ (note that the base is the largest number and can therefore be x or y). It's easy to see that $\log_x(z)$ converges to n as $\Delta Q \rightarrow \pm\infty$, where ΔQ is the difference between x and y . I am certain that $n/\log_x(z)$ is never > 1 since the smallest ΔQ is 0 when $x = y$ with which $n/\log_x(z)$ doesn't exceed 1. Because of this we can introduce the floor function to get an integer solution,

$$n = \lfloor \log_x(z) \rfloor \tag{1}$$

1.1.1 Example

Consider the following equation,

$$10^n + 9^n = 1729$$

Using the (1),

$$n = \lfloor \log_x(z) \rfloor \\ \lfloor \log_{10}(1729) \rfloor$$

which is equal to 3. Plugging it back in to check...

$$10^3 + 9^3 = 1729$$

1.1.2 Counterexample

Consider,

$$2^n + 2^n = 8$$

Using the formula to find n

$$\lfloor \log_2(8) \rfloor = 3$$

$$2^3 + 2^3 \neq 8$$

1.2 Solutions To The Equation When $x = y = 2$

The reason $n = \lfloor \log_x(z) \rfloor$ doesn't work in the counterexample is due to the property 2 has where $2 + 2 = 2 \times 2 = 2^2$. For any equation where $x = y = 2$, $n + 1$ will always equal $\lfloor \log_2(z) \rfloor$ keeping in mind that $x, y, z, n \in \mathbb{N}$. Therefore we can say that

$$n + 1 = \lfloor \log_2(z) \rfloor$$

for an equation of the form

$$2^n + 2^n = z$$